

Potential Benefits of Isabelle/HOL Integration

Isabelle/HOL represents one of the most mature and academically established interactive theorem provers available. Its integration into the system would provide a high level of formal assurance and theoretical robustness.

First, Isabelle's core inference kernel is small and rigorously verified, ensuring that any derived theorem is logically sound with respect to higher-order logic. Unlike a handcrafted rule engine, which may contain subtle implementation errors, Isabelle guarantees correctness at the foundational level. Integrating Isabelle would therefore elevate the system from structured rule validation to formally certified proof verification.

Second, Isabelle already includes formalised natural deduction frameworks, most notably through projects such as NaDeA (Natural Deduction Assistant). This provides a directly relevant formal model upon which the implemented system is conceptually based. Leveraging Isabelle would align the educational interface with a formally verified logical specification rather than an independently implemented approximation.

Third, Isabelle is widely used in formal methods research and industrial verification contexts. It has been applied in:

- Verified compiler construction
- Security protocol verification
- Formal verification of mathematical theorems
- Proofs of correctness in safety-critical systems

Exposing students to a system backed by Isabelle would therefore connect introductory natural deduction exercises with real-world formal verification technologies.

Another significant advantage is extensibility. Isabelle/HOL supports:

- Propositional logic
- First-order logic
- Higher-order logic
- Inductive definitions
- Formal reasoning about programs

This would allow the educational platform to scale beyond propositional natural deduction and potentially support richer logical systems or even formal program verification in future iterations.

Finally, integrating Isabelle would increase the academic credibility of the system. A formal backend verification layer demonstrates rigorous alignment between the implemented system and established formal logic theory. This strengthens both the theoretical foundation and the long-term sustainability of the platform.